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Review

Source attribution studies of foodborne pathogens, 2010–2023: a review and collection of estimates



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ABSTRACT

Identifying the most important sources and transmission routes of foodborne pathogens is crucial for developing food safety strategies at both national and regional levels. Various source attribution approaches and methods have been applied worldwide. This review aims to provide an overview of available studies that estimate the attribution of foodborne illnesses globally, regionally, and nationally, compiled by pathogen and region. We conducted a search in PubMed to identify peer-reviewed source attribution studies of one or more foodborne pathogens published from January 2010 to July 2023. Additionally, we consulted experts of the Foodborne Disease Burden Epidemiology Reference Group (FERG) for 2021–2025 and screened reference lists of included articles to identify further relevant studies. The extracted studies were categorized by pathogen, source attribution method, country, and region. We identified 62 studies published during the specified period, covering 34 pathogens across 26 countries, and extracted source attribution estimates. The most frequently studied pathogens were *Salmonella*, *Campylobacter*, and *Escherichia coli*. The methods most used were microbial subtyping, outbreak analysis, and mixed methods. Extracted data show substantial variation in results across methods and countries, highlighting the challenges of reaching a consensus on the relative contributions of foodborne disease sources at both regional and global levels based on empirical data. All extracted estimates are available in supplementary materials.

1. Introduction

Foodborne diseases can be caused by a range of hazards, including bacteria, viruses, parasites, and chemical contaminants. Identifying the most important sources of foodborne pathogens is essential to define food safety strategies at national and regional levels ((Hoffmann et al. (2021), FAO (2006))). The sources and relative contribution of different foods and transmission routes to the burden of foodborne diseases may vary by region and by country. Thus, estimating attribution proportions using local data is important to complement estimates of the burden of foodborne diseases and identify food safety priorities.

Foodborne illness source attribution studies estimate the relative contributions of various sources or transmission routes to the burden of foodborne diseases (Pires et al. (2009)). These methods can attribute foodborne illnesses at different stages of the supply chain: from the pathogen reservoir (the origin of the pathogen), through food processing and distribution, to the point of human exposure to contaminated food. A variety of source attribution methods are available, including microbiological approaches, epidemiological approaches, expert elicitation, and intervention studies. Microbiological methods can analyze microbial data from on-farm testing, public health surveillance, food inspections, and various food processing and handling monitoring

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programs. Meanwhile, epidemiological methods focus on analyzing data from outbreak investigations and sporadic disease surveillance (case-control studies, cohort studies, time-series analysis). The results of different methods can be useful to answer different One Health questions.

In recent years, numerous source attribution studies for foodborne pathogens have been conducted globally, employing different methods depending on the specific hazard, data availability, and the capacity of each country. A review of the estimates from these studies is necessary to facilitate the analysis of the relative contributions of various sources of foodborne infections across different countries and regions, compare results from different methods, and identify gaps in source attribution estimates for specific pathogen-region-method combinations. This review aims to address that need.

We reviewed global, regional, and national foodborne illness source attribution studies. This study was commissioned by the World Health Organization (WHO) to inform their initiative to estimate the global burden of foodborne diseases. This work was then overseen by the WHO Foodborne Disease Burden Epidemiology Reference Group (FERG) for 2021–2025 (WHO, FERG).

2. Methods

2.1. Literature search and selection of studies

We conducted a search using PubMed to identify peer-reviewed studies published between January 2010 and July 2023 that estimated the relative contributions of various transmission routes for foodborne infections, as well as the relative contributions of different foods as exposures. The transmission routes included food, water, human contact, animal contact, and airborne transmission. We searched the literature using different combinations of the terms: “source attribution”, “foodborne disease”, “foodborne pathogens”, “food source”, and “human OR human samples”. No language restrictions were applied to the search. The full search strategies can be found in Table S1 (Supplementary File). We conducted additional consultations with experts from the FERG for 2021–2025 to ensure that no relevant studies were overlooked during the formal search. Finally, we reviewed the reference lists of the included articles to identify any further relevant studies.

The search covered the pathogens prioritized by the FERG for 2021–2025 Enteric Disease Task Force and Parasitic Disease Task Force by using a method of expert elicitation. The full list of the pathogens is shown in Table S2 (Supplementary File). If studies reported source attribution estimates for other pathogens, those were also considered. Only studies reporting source attribution estimates, such as proportion of attributed cases or equivalent, were included.

Unrelated studies that were accidentally captured by the search string, including outbreak reports, studies without new information (reviews), studies conducted on data sets older than 2010, studies that investigated only waterborne infections, and studies that reported results from food sample testing, were excluded.

The initial eligibility for inclusion was carried out by one reviewer, who screened titles, abstracts, and when necessary, the full texts. A further full-text screening was conducted on all articles selected during the initial assessment. Articles that did not meet the inclusion criteria but could provide some useful information, such as relevant references, were categorized as “Other”.

From the selected articles, we extracted data on the following variables: bibliographic reference of the study, DOI, title, year of publication, source attribution approach, country and WHO region where data were collected (i.e., African Region, Region of Americas, European Region, Eastern Mediterranean Region, South-East Asia Region, and Western Pacific Region) (WHO, World Health Statistics), time period of data collection, pathogen, data sources, method for uncertainty assessment (if relevant), type of source attribution estimate, and values of the source attribution estimates for each source and transmission pathway.

To make estimates comparable, the different types of source attribution estimates were standardized to percent of attributed cases (%). For example, if a study reported a fraction (e.g., 0.1) we converted them to 10 %. We categorized the studies according to the six WHO regions based on the country or region where each study was conducted.

2.2. Data extraction and categorization

To harmonize the variety of transmission routes, food sources described in the included studies were categorized using an adaptation of the hierarchical food categorization scheme from Painter et al. (2009). If data were available for food subcategories, these were preserved, and all available data were extracted. We categorized source attribution studies by source attribution method applied as defined by Batz et al. (2005) and Pires et al. (2009). We classified methods used as: microbial subtyping, comparative exposure assessment, analysis of data from outbreak investigations or active surveillance, systematic reviews of case-control studies, expert elicitations, and intervention studies, or mixed methods. We also extracted data on the analytical approach used and on whether an uncertainty analysis was conducted (yes/no).

3. Results

Of the 1927 records initially identified through the PubMed MEDLINE database, 45 were flagged for inclusion in this review. Additionally, we included another seventeen articles identified through FERG for 2021–2025 expert consultations. The total number of articles included in the review was 62. Fig. 1 shows the flow diagram of data screening and study selection process.

3.1. Geographic representation

The 62 included studies were conducted in 26 countries, globally, or regionally (Tables 1 and 2, S3 (Supplementary File)). Twenty-nine studies were conducted in the European region, specifically in Austria, Denmark, Estonia, Finland, France, Germany, Italy, Latvia, Lithuania, Luxembourg, Netherlands (Kingdom of the), Slovenia, Switzerland, and the United Kingdom of Great Britain and Northern Ireland. Two of those 29 studies were conducted at the regional level. One study was carried out in the WHO African region, covering three countries: Burkina Faso, Ethiopia, and Rwanda. From the Western Pacific Region, eleven studies from five countries (Australia, Cambodia, Japan, China, and New Zealand) and one at the WHO Western Pacific regional level were identified. In the Region of the Americas, 14 studies from three countries (Brazil, Canada, and United States of America), one regional-level study covering Latin America and the Caribbean, and one study covering the whole WHO Region of the Americas regional level were identified. One study was conducted in the WHO Eastern Mediterranean region (in Egypt). Six studies were conducted globally. Of these, two studies reported estimates at the WHO subregional level. These studies were the only ones providing estimates for the WHO South-East Asian region.

3.2. Source attribution approaches used

Source attribution studies using the following approaches were identified: microbial subtyping, comparative exposure (risk) assessment, systematic review and meta-analysis of cross-sectional and case-control studies, analyses of data from outbreak investigations or active surveillance, expert elicitation, intervention studies, or a combination of methods (Table S3, Supplementary File).

Microbial subtyping was most frequently applied (43 out of 62 studies), but only for four pathogens: *Salmonella* spp., *Listeria monocytogenes*, *Campylobacter* spp., and *E. coli* (Shiga toxin-producing *E. coli* (STEC) and extended-spectrum beta-lactamases (ESBL) -producing *E. coli*) (Table S3, Supplementary File). Among the included studies that applied microbial subtyping approaches for source attribution,

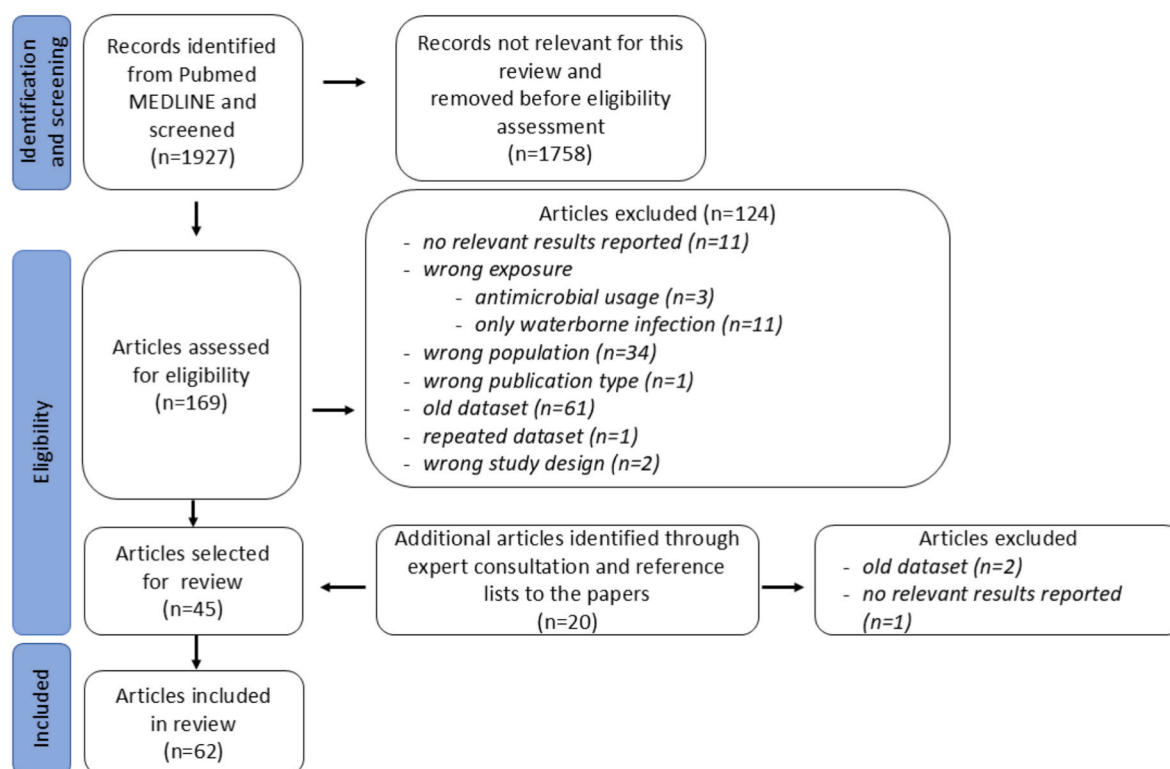


Fig. 1. Flow chart of the selection process of source attribution studies of foodborne pathogens published in 2010–2023.

pathogen subtyping data also varied. Nineteen studies (45 %) applied Multi-Locus Sequence Typing (MLST); 20 % applied serotyping and phage typing; 10 % applied PCR-based methods, such as RT-PCR, and 7 % applied Multi-Locus Variable Number Tandem repeat (MLVA) (<https://doi.org/10.5281/zenodo.14720777>). The analytical models applied included the Hald model (13 %), STRUCTURE (28 %) the Asymmetric Island model (AIM) (12 %), the Dutch model (7 %), or modifications thereof (16 %). 7 % of these studies applied machine-learning models. Eight studies analyzed data from outbreak investigations for source attribution of 12 pathogens: STEC, EHEC, *E. coli*, *Salmonella* (non-typhoidal), *Campylobacter* sp., *Vibrio parahaemolyticus*, *Staphylococcus aureus*, *Clostridium perfringens*, *Shigella* sp., *Bacillus cereus*, *Listeria monocytogenes*, and norovirus. One study conducted a systematic review of case-control and cross-sectional studies of *Echinococcus* sp. Expert elicitation was applied to a broader variety of pathogens at both country and global level. Twelve studies used mixed methods for source attribution, such as microbial subtyping combined with case-control studies, expert elicitation, exposure assessment, or machine-learning. Three of these studies represent the combination of microbial subtyping methods and case-control studies for source attribution. Mughini-Gras et al. (2022) combined expert elicitation, microbial subtyping and case-control studies to attribute the infections caused by non-typhoidal *Salmonella*, *Listeria monocytogenes*, STEC, and *Campylobacter* sp. Two studies combined microbial subtyping with exposure assessment.

3.3. Pathogens and transmission routes

Source attribution data were extracted for 34 pathogens, including bacteria, parasites, and viruses (Fig. 2, Table 1). Fig. 2 shows an overview of the proportion of source attribution studies conducted by pathogen relative to all studies identified, including all settings and approaches. The pathogens most frequently included in studies were *Campylobacter* spp. (30 studies), *Salmonella* spp. (in 29 studies), and *E. coli* (22 studies). The range of sources included in studies varied

substantially across studies and across pathogens. The general transmission pathways covered in the range of studies included food, water, animal contact, environment, and human-to-human. The spectrum of identified sources is presented in Table 1.

3.4. Reporting of estimates, data sources, and analytical models applied

Source attribution studies also differed in the way they reported the estimates (e.g., attribution proportions, fractions), how they addressed statistical uncertainties, and the data sources used for source attribution (Table 2). All source attribution estimates extracted from studies are available in through <https://doi.org/10.5281/zenodo.14720777>, where the data were organized by WHO region and group of pathogens.

3.5. Overview of source attribution estimates by pathogen

All extracted source attribution estimates by pathogen and WHO-region are available through <https://doi.org/10.5281/zenodo.14720777>. Despite the differences in the number of included sources, methods applied, and data used, estimates for the same pathogens in some regions are largely consistent in terms of the ranking of the sources contributing most to disease. For example, most collected studies on *Campylobacter jejuni* in the EUR estimated that the top sources of infection are poultry (including broilers), cattle, small ruminants and pigs; in the AMR that the top sources are cattle and broilers; in the AFR broilers; and in the SEAR dairy and broilers. Although estimates agreed in terms of ranking, source attribution proportions varied considerably. The top sources of *Salmonella* spp. were broilers, eggs, and pigs in all regions, but ranked differently across regions. Several studies in the region of Americas estimated attribution proportions for *E. coli* (other than ETEC) (to the food, water, animal contact, environmental pathways, and human-to-human transmission), with a noticeable predominance of water as a source of the infection. The food sources of non-ETEC *E. coli* infection was meat, goat milk, dairy, seafood, oils and sugars, grains and beans, poultry, and produce. The main sources of ETEC were

Table 1

Transmission routes and food sources for different pathogens and countries extracted from source attribution studies, 2010–2023.

Pathogen	Number of countries/WHO regions where source attribution was conducted	Transmission pathways	Food sources	References
Bacteria				
<i>E. coli (other than ETEC)</i>	2 countries (Canada, United States of America (USA)) (AMR), 1 regional study (Latin America and the Caribbean) (AMR), 1 global study	Food, water, animal contact, environment, human-to-human	Beef, meat, goat milk, dairy, seafood, oils and sugars, grains and beans, broilers, poultry, ducks, fruits and nuts, produce, mixed and unspecified food	Batz et al. (2021), Beshearse et al. (2021), Butler et al. (2015), CDC, 2023, Pires et al. (2012)
<i>EHEC</i>	1 country (Japan) (WPR)	Food	Beef, horse, broilers, vegetables	Kumagai et al. (2020)
<i>EPEC</i>	1 global study (AFR, AMR, EUR, EMR, SEAR, WPR)	Water, food, air, human-to-human	Mixed and unspecified food	Hald et al. (2016)
<i>ESBL-producing E. coli</i>	2 countries (Netherlands (Kingdom of the), Germany) (EUR)	Food, companion animals, dogs, environment (farm), high risk groups, human-to-human, unknown	Beef, pig, horse, cattle, small ruminants, seafood, broilers, turkey, game, vegetables	Mughini-Gras et al. (2019a,b), Perestrelo et al. (2022)
<i>ETEC</i>	2 countries (Ethiopia, Burkina Faso (AFR)), 1 global study	Food, water, air, human-to-human, animal contact	Beef, ruminants, dairy, oils and sugars, grains and beans, poultry, fruits and nuts, vegetables, mixed and unspecified food	Hald et al. (2016), Sapp et al. (2022)
<i>ETEC</i>	1 global study (WPR, SEAR, EUR)	Water, air, food, human-to-human, animal contact	Mixed and unspecified food	Hald et al. (2016)
<i>ETEC</i>	2 countries (Canada, USA) (AMR), 1 global study	Food, water, animal contact, environment, human-to-human	Mixed and unspecified food	Beshearse et al. (2021), Butler et al. (2015), Hald et al. (2016)
<i>Bacillus cereus</i>	1 country (USA) (AMR), 1 regional study (Latin America and the Caribbean) (AMR)	Food	Meat, mutton, dairy, seafood, grains and beans, broilers, poultry, produce, mixed and unspecified food	Butler et al. (2015), Pires et al. (2012)
<i>Brucella sp.</i>	1 country (Rwanda) (AFR)	Food	Beef, pig, small ruminants, dairy, mixed and unspecified food	Sapp et al. (2022)
<i>Brucella sp.</i>	2 global studies	Food, water, animal contact, environment, human-to-human	Mixed and unspecified food	Hald et al. (2016), Hoffmann et al. (2017)
<i>Brucella sp.</i>	2 countries (Canada, USA) (AMR)	Food, water, animal contact, environment, human-to-human	Mixed and unspecified food	Beshearse et al. (2021), Butler et al. (2015)
<i>Campylobacter spp.</i>	2 countries (Canada, USA) (AMR)	Food, water, animal contact, environment (farm, water), human-to-human	Beef, dairy, fish, seafood, oils and sugars, broilers, turkey, game, poultry, fruits, seeded vegetables, vegetables, row crops vegetables, produce, cattle, pig, leafy greens, frozen fruits and vegetables, fruits eaten without peeling, berries, fruits eaten peeled, juice, whole meat, ground beef, ground meat (other than beef), frozen chicken, fresh chicken, turkey, seafood/fish, nuts and seeds, snacks, mixed and unspecified food, foods by preparation method	Batz et al. (2012), Beshearse et al. (2021), Butler et al. (2015), Hsu et al. (2020), Hudson et al. (2021), Pintar et al. (2017), Ravel et al. (2017), Teixeira et al. (2022))
<i>Campylobacter sp.</i>	3 countries (Ethiopia, Rwanda, Burkina Faso) (AFR)	Food	Beef, pig, small ruminants, dairy, broilers, fruits and nuts, vegetables, mixed and unspecified food	Sapp et al. (2022)
<i>Campylobacter sp.</i>	1 country (Egypt) (EMR)	Food	Dairy, broilers	Saif et al. (2022)
<i>Campylobacter spp.</i>	13 countries (Italy, Scotland (United Kingdom of Great Britain and Northern Ireland), Denmark, Latvia, Estonia, Lithuania, Switzerland, Slovenia, Austria, Germany, Luxembourg, Netherlands (Kingdom of the), France) (EUR)	Food, environment (including water), dogs, companion animals	Pig, cattle, small ruminant, ruminant, sheep, broilers, poultry, pheasants, game, turkey, ducks, beef, calves, raw milk (urban), vegetables	Berthenet et al. (2019), Di Giannatale et al. (2016), Kittl et al. (2013), Kovac et al. (2018), Mäesaar et al. (2020), Mossong et al. (2016), Mughini-Gras et al. (2021), Pires and Christensen, 2017; Rosner et al. (2017), Seguino et al. (2018), Thépault et al. (2018), Wainaina et al. (2022)
<i>Campylobacter spp.</i>	4 global studies	Food, animal contact, dogs, reptiles, environment	Cattle, mutton, red meat, pig, small ruminant, broilers, poultry, turkey, laying hens	Arning et al. (2021), Hald et al. (2016), Hoffmann et al. (2017), Mughini-Gras et al. (2022)
<i>Campylobacter spp.</i>	3 countries (Japan, New Zealand, Australia) (WPR)	Food, water, environment (water), unknown	Beef, meat, dairy, fish, broilers, turkey, ducks, game, small ruminant, ruminant, poultry, cattle, pig	Kumagai et al. (2020), Lake et al. (2021), McLure et al. (2023), Nohra et al. (2016), (2020); Osbjør et al. (2016)
<i>Clostridium spp.</i>	1 country (Canada) (AMR), 1 regional study (Latin America and the Caribbean) (AMR)	Food, water, animal contact, human-to-human	Beef, meat, goat milk, dairy, seafood, oils and sugars, grains and beans, broilers, poultry, ducks, fruits and nuts, produce, mixed and unspecified food	Butler et al. (2015), Pires et al. (2012)
<i>Clostridium perfringens</i>	1 country (Japan) (WPR)	Food	Beef, shellfish, grains and beans, broilers, ducks, vegetables	Kumagai et al. (2020)

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Table 1 (continued)

Pathogen	Number of countries/WHO regions where source attribution was conducted	Transmission pathways	Food sources	References
<i>Legionella</i> sp.	1 country (USA) (AMR)	Water		Beshearse et al. (2021)
<i>Listeria monocytogenes</i>	2 countries (USA, Canada) (AMR)	Food, water, animal contact, human-to-human	Beef, pig, dairy, fish, broilers, turkey, eggs, fruits, vegetables row crops, produce, seeded vegetables, sprouts	Batz et al. (2012), Butler et al. (2015), CDC (2023)
<i>Listeria monocytogenes</i>	2 countries (Italy, Denmark) (EUR)	Food	Beef, pig, cattle, dairy, fish, broilers, poultry, ducks, game, laying hens, mixed and unspecified food	Filipello et al. (2020), Merlotti et al. (2020)
<i>Listeria monocytogenes</i>	N/A	Food, human-to-human, unknown	Meat, dairy, seafood, poultry, fruits, leafy green vegetables, vegetables, pig, cattle, small ruminant, fish, broilers, turkey	Mughini-Gras et al. (2022), Tanui et al. (2022)
<i>Mycobacterium</i> sp.	1 country (Rwanda) (AFR)	Food	Dairy	Sapp et al. (2022)
<i>Mycobacterium</i> sp.	1 country (USA) (AMR)	Food, water, animal contact, environment, human-to-human	Mixed and unspecified food	Beshearse et al. (2021)
<i>Pseudomonas</i> sp.	1 country (USA) (AMR)	Water, animal contact, environment,		Beshearse et al. (2021)
<i>Salmonella</i> sp.	3 (Ethiopia, Rwanda, Burkina Faso) (AFR)	Food	Beef, pig, small ruminant, dairy, fish, shellfish, oils and sugars, grains and beans, broilers, eggs, fruits and nuts, vegetables, mixed and unspecified food	Sapp et al. (2022)
<i>Salmonella</i> sp.	1 country (USA) (AMR), 1 regional study (Latin America and the Caribbean) (AMR)	Food, water, animal contact (including companion animal), environment (including farm and water), human-to-human	Meat, dairy, bakery, broilers, eggs, vegetables, beef, pig, goat milk, fish, seafood, oils and sugars, vegetables row crops, produce, vegetables and fruits, fruits, fruits and nuts, nuts, seeded vegetables, sprouts, grains and beans, poultry, turkey, duck, game, mixed and unspecified food	Ahlstrom et al. (2017), Batz et al. (2012), Beshearse et al. (2021), Butler et al. (2015), CDC (2023), Christidis et al. (2020), Neto et al. (2021), Pires et al. (2012), Sher et al. (2021), Vieira et al. (2016)
<i>Salmonella</i> sp.	6 countries (Netherlands (Kingdom of the), United Kingdom of Great Britain and Northern Ireland, Germany, Italy, Finland, Denmark) (EUR)	Food, water, animal contact (mammals, dogs, reptiles), environment (including animal feed and farm), unknown	Red meat, beef, meat, sheep, dairy, cheese, fish and seafood, grains and beans, broilers, turkey, eggs, herbs and spices, produce, pig, cattle, ruminant, small ruminant, laying hens, poultry, ducks, mixed and unspecified	Arnold et al. (2021), Barco et al. (2015), Chanamé Pinedo et al. (2022), Jabin et al. (2019), Merlotti et al. (2020), Mikkilä et al. (2019), Mughini-Gras et al. (2014a,b), Mughini-Gras et al. (2016), (2020); Munck et al. (2020)
<i>Salmonella</i> spp.	3 global studies	Food, animal contact (reptiles)	Pig, cattle, broilers, poultry, laying hens, eggs	Hald et al. (2016), Hoffmann et al. (2017), Mughini-Gras et al. (2022)
<i>Salmonella</i> spp.	2 countries (Japan, Australia) (WPR)	Food	Beef, sheep, fish and seafood, broilers, ducks, game, eggs, vegetables, small ruminant, nuts, laying hens	Fearnley et al. (2018), Glass et al. (2016), Kumagai et al. (2020), McLure et al. (2022)
<i>Shigella</i> spp.	1 global study	Water, air, food, soil, human-to-human		Hald et al. (2016)
<i>Shigella</i> spp.	2 countries (Canada, USA) (AMR), 1 regional study (Latin America and the Caribbean) (AMR)	Food, water, animal contact, environment, human-to-human	Meat, goat milk, dairy, seafood, oils and sugars, grains and beans, broilers, poultry, eggs, mixed and unspecified food	Beshearse et al. (2021), Butler et al. (2015), Pires et al. (2012)
<i>Staphylococcus aureus</i>	2 countries (Canada, USA) (AMR), 1 regional study (Latin America and the Caribbean) (AMR)	Food, water, animal contact, environment, human-to-human	Beef, meat, mutton, sheep, dairy, seafood, oils and sugars, grains and beans, broilers, poultry, game, eggs, fruits and nuts, produce, mixed and unspecified food	Beshearse et al. (2021), Butler et al. (2015), Pires et al. (2012)
<i>Staphylococcus aureus</i>	1 country (Japan) (WPR)	Food, water	Beef, dairy, fish, seafood, grains and beans, broilers, eggs, vegetables	Kumagai et al. (2020)
<i>STEC</i>	1 country (Ethiopia) (AFR)	Food	Beef, small ruminant, dairy, fruits and nuts, vegetables, mixed and unspecified food	Sapp et al. (2022)
<i>STEC</i>	1 country (USA) (AMR), 1 regional study (AMR)	Food, water, animal contact, environment, human-to-human, unknown	Beef, meat, sheep, dairy, seafood, produce, pig, fish, turkey, grains and beans, nuts, broilers, game, eggs, fruits, seeded vegetables, vegetables row crops, sprouts, mixed and unspecified food	Batz et al. (2012), Beshearse et al. (2021), CDC (2023), Pires et al. (2019)
<i>STEC</i>	1 country (Netherlands (Kingdom of the)) (EUR), 1 regional study (EUR)	Food, unknown	Beef, meat, pig, sheep, cattle, small ruminant, dairy, seafood, grains and beans, broilers, game, poultry, eggs, produce	Mughini-Gras et al. (2018), Pires et al. (2019)
<i>STEC</i>	3 global studies	Food	Red meat, pig, cattle, broilers	Hald et al. (2016), Hoffmann et al. (2017), Mughini-Gras et al. (2022)
<i>STEC</i>	1 regional study (WPR)	Food, unknown	Beef, meat, dairy, grains and beans, game, produce	Pires et al. (2019)

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Table 1 (continued)

Pathogen	Number of countries/WHO regions where source attribution was conducted	Transmission pathways	Food sources	References
<i>Streptococcus</i> spp.	1 country (USA) (AMR)	Food, water, animal contact, environment, human-to-human	Mixed and unspecified food	Beshearse et al. (2021)
<i>Vibrio</i> spp.	2 countries (USA, Canada) (AMR)	Food, water, animal contact, human-to-human	Mixed and unspecified food	Beshearse et al. (2021), Butler et al. (2015)
<i>Vibrio parahaemolyticus</i> .	1 country (Japan) (WPR)	Food	Fish, shellfish	Kumagai et al. (2020)
<i>Vibrio cholerae</i>	1 global study	Water, air, food, human-to-human		Hald et al. (2016)
<i>Yersinia enterocolitica</i>	2 countries (USA, Canada) (AMR)	Food, water, animal contact, human-to-human	Mixed and unspecified food	Beshearse et al. (2021), Butler et al. (2015)
Parasites				
<i>Alveolar Echinococcus</i>	3 global study	Food, animal contact (fox, dog), environment	Mixed and unspecified food, fruits, vegetables	Hald et al. (2016), Hoffmann et al. (2017), Torgerson et al. (2020)
<i>Ascaris</i> sp.	2 global study	Food, water, animal contact, environment, human-to-human	Mixed and unspecified food	Hald et al. (2016), Hoffmann et al. (2017)
<i>Cryptosporidium</i> spp.	1 country (Rwanda) (AFR), 2 global studies	Food, water, animal contact, human-to-human	Dairy, fruits and nuts, vegetables, mixed and unspecified food	Hald et al. (2016), Hoffmann et al. (2017), Sapp et al. (2022)
<i>Cryptosporidium</i> spp.	2 global studies	Food, water, animal contact, human-to-human	Mixed and unspecified food	Hald et al. (2016), Hoffmann et al. (2017)
<i>Cryptosporidium</i> spp.	1 country (Canada) (AMR), 2 global studies	Food, water, animal contact, human-to-human	Dairy, fruits and nuts, vegetables, mixed and unspecified food	Butler et al. (2015), Hald et al. (2016), Hoffmann et al. (2017)
<i>Cyclospora</i> spp.	1 country (Canada) (AMR)	Food, water, animal contact, human-to-human	Mixed and unspecified food	Butler et al. (2015)
<i>Cystic echinococcus</i>	1 global study	Food, animal contact (dog), environment (water)	Mixed and unspecified food	Torgerson et al. (2020)
<i>Entamoeba histolytica</i>	2 global studies	Food, other	Fruits, vegetables	Hald et al. (2016), Hoffmann et al. (2017)
<i>Giardia</i> spp.	1 country (Canada) (AMR), 1 global study	Food, water, animal contact, human-to-human	Mixed and unspecified food	Butler et al. (2015), Hald et al. (2016)
<i>Giardia</i> spp.	1 global study (EMR, EUR, WPR, SEAR, AMR)	Food, water, animal contact, human-to-human		Hald et al. (2016)
<i>Toxoplasma gondii</i>	1 country (Canada) (AMR)	Food, water, animal contact, human-to-human	Mixed and unspecified food	Butler et al. (2015)
<i>Toxoplasma gondii</i>	2 global studies	Water, food, soil	Dairy, fruits, vegetables, beef, small ruminants, pork, poultry, eggs	Hald et al. (2016), Hoffmann et al. (2017)
<i>Toxoplasma gondii</i>	1 country (China) (WPR)	Food	Pig, cattle, small ruminant, broilers, ducks	Deng et al. (2020)
<i>Trichinella</i> spp.	1 country (Canada) (AMR)	Food, animal contact	Mixed and unspecified food	Butler et al. (2015)
Viruses				
Hepatitis A	1 country (Canada) (AMR)	Food, water, animal contact, human-to-human	Mixed and unspecified food	Butler et al. (2015)
Hepatitis A	1 global study	Food, water, animal contact, human-to-human, soil		Hald et al. (2016)
Norovirus	1 country (Canada) (AMR)	Food	Beef, seaweed, fish, shellfish, grains and beans, broilers, fruits and nuts, vegetables	Butler et al. (2015)
Norovirus	1 global study	Food, water, animal contact, human-to-human, soil	Mixed and unspecified food	Hald et al. (2016)
Norovirus	1 country (Japan) (WPR)	Food, water, animal contact, human-to-human	Mixed and unspecified food	Kumagai et al. (2020)
Rotavirus	1 country (Canada) (AMR)	Food, water, animal contact, human-to-human	Mixed and unspecified food	Butler et al. (2015)

*AFR: African Region. AMR: Region of the Americas. EMR: Eastern Mediterranean Region. EUR: European Region. SEAR South-East Asia Region. WPR: Western Pacific Region.

dairy and poultry in the AFR and mixed and specified food in WPR, SEAR, EUR, and AMR (<https://doi.org/10.5281/zenodo.14720777>; Table SF2). Most studies identified beef, small ruminants, and cattle (AFR, AMR and EUR) as the most important sources of STEC (<https://doi.org/10.5281/zenodo.14720777>; Table SF2).

There are also studies reporting estimates that lead to different rankings. These differences apply to all pathogens and regions and may reflect the method applied, data used, sources included, or time period (<https://doi.org/10.5281/zenodo.14720777>).

4. Discussion

Interest in foodborne illness source attribution has grown globally

over the years. Our review identified 62 source attribution studies published between 2010 and 2023, conducted in 26 countries across six WHO regions, as well as at regional or global levels. These studies estimated the relative contributions of various sources to human diseases caused by different foodborne pathogens, including bacteria, viruses, and parasites. While most studies reported national estimates, some focused on regional or global assessments. The studies varied in source attribution methods, analytical models, and the types and representativeness of data used. The most frequently applied method was microbial subtyping, particularly for source attribution of *Campylobacter* spp., *Salmonella* spp., and *E. coli*.

Although the 62 studies covered the six WHO regions and a variety of pathogens, they show substantial gaps in attribution estimates for many

Table 2

Description of source attribution studies of foodborne pathogens, 2010–2023: target population, pathogens included, data sources, period of data collection, and reporting of uncertainties.

Author (year)	Target population	Pathogen	Data sources (method)	Time period for data collection	Statistical uncertainty provided
Pires et al. (2019)	AMR, EUR, WPR	<i>STEC</i>	Outbreak surveillance, publicly available reports and datasets, and grey literature reports	1998–2017	95 % uncertainty interval
Chanamé Pinedo et al. (2022)	EUR	<i>Salmonella</i> spp.	Outbreak surveillance (EFSA)	2015–2019	95 % uncertainty interval
Sher et al. (2021)	USA	<i>Salmonella</i> spp.	Outbreak surveillance CDC Foodborne Disease Outbreak Surveillance System	1990–2015	No
Kumagai et al. (2020)	Japan	<i>Campylobacter</i> sp., <i>EHEC</i> , <i>Salmonella</i> spp., <i>Vibrio parahaemolyticus</i> , <i>Clostridium perfringens</i> , norovirus	Outbreak surveillance (compiled by the Ministry of Health, Labour and Welfare, data of outbreak investigations undertaken by local governments, and infectious disease surveillance data on <i>EHEC</i> and norovirus)	2007–2018	95 % uncertainty interval
Pires et al. (2012)	Latin America and Caribbeans (Bahamas, Mexico, Argentina, Chile)	<i>Salmonella</i> spp., <i>E. coli</i> , <i>Clostridium perfringens</i> , <i>Staphylococcus aureus</i> , <i>Bacillus cereus</i>	Outbreak surveillance (SIRVETA)	1993–2010	95 % uncertainty interval
Batz et al. (2021)	USA	<i>Salmonella</i> spp., <i>STEC</i> , <i>Campylobacter</i> sp., <i>Listeria monocytogenes</i>	Outbreak surveillance (CDC Foodborne Disease Outbreak Surveillance System)	1998–2012	95 % uncertainty interval
CDC (2023)	USA	<i>Salmonella</i> spp., <i>STEC</i> , <i>Listeria monocytogenes</i>	Outbreak surveillance (CDC Foodborne Disease Outbreak Surveillance System)	1998–2012	No
Neto et al. (2021)	Brazil	<i>Salmonella</i> spp.	Outbreak surveillance. 2011–2015: sporadic cases captured by Hospital Epidemiological Surveillance Nuclei (NVEH) from Global Salm-Surv (GSS)-linked hospitals (8 out of 19 Regional Health Coordination (CRS) from the sub-regions of RS). 2005 to 2015: outbreak data	2005–2015	95 % uncertainty interval
Glass et al. (2016)	Australia	<i>Salmonella</i> spp.	Notifiable Infectious Diseases Surveillance Database	2000–2010	95 % uncertainty interval
Filipello et al. (2020)	Italy	<i>Listeria monocytogenes</i>	Voluntary network of hospital laboratories in two Northern Italy regions, i.e. Lombardy and Piedmont (clinical samples). Routine surveillance by the Regional Animal Health and Food Safety Institutes (IZS) and Department of Veterinary Sciences of the University of Turin (food isolates)	2004–2016	95 % uncertainty interval
Fearnley et al. (2018)	Australia	<i>Salmonella</i> spp.	Communicable disease surveillance system (by Queensland Department of Health)	2000–2011	95 % uncertainty interval
Di Giannatale et al. (2016)	Italy	<i>Campylobacter</i> sp.	Active surveillance	2012	No
Seguino et al. (2018)	Scotland (United Kingdom of Great Britain and Northern Ireland)	<i>Campylobacter</i> spp.	Collection of caecal samples from pheasant carcasses (sampling from 5 geographical regions in Scotland) during the study; PubMLST database	2013–2014	No
Nohra et al. (2016)	New Zealand	<i>Campylobacter</i> sp.	Molecular Epidemiology and Public Health Laboratory of Massey University; monitoring programs for suppliers; PubMLST database	2005–2014	95 % uncertainty interval
Vieira et al. (2016)	USA	<i>Salmonella</i> sp.	National Antimicrobial Resistance Monitoring System; State public health laboratories	1996–2012	95 % uncertainty interval
Mughini-Gras et al. (2014a, b)	Italy	<i>Salmonella</i> sp.	Enter-net Italia; SIMI	2002–2010	95 % uncertainty interval
Wainaina et al. (2022)	Denmark	<i>Campylobacter</i> sp.	National public health surveillance, Food monitoring, Public Health England (PHE)	2015–2017	No
Arnold et al. (2021)	United Kingdom of Great Britain and Northern Ireland	<i>Salmonella</i> sp.	National public health surveillance/monitoring/control programs/research projects, Public Health England (PHE)	2014–2016	No
Mäesaar et al. (2020)	Latvia, Lithuania, Estonia	<i>Campylobacter</i> sp.	Published dataset	2008–2016	95 % uncertainty interval
Mughini-Gras et al. (2018)	Netherlands (Kingdom of the)	<i>STEC</i>	Dutch National Institute for Public Health and the Environment (RIVM); Netherlands' Food and Consumer Product Safety Authority (NVWA) as part of national surveillance activities	2010–2014	95 % uncertainty interval

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Table 2 (continued)

Author (year)	Target population	Pathogen	Data sources (method)	Time period for data collection	Statistical uncertainty provided
Kittl et al. (2013)	Switzerland	<i>Campylobacter</i> sp.	Slaughterhouses in the framework of monitoring programs, fecal samples submitted to in-house diagnostics department, Swiss National Centre for Enteropathogenic Bacteria (NENT)	2002–2012	No
Kovac et al. (2018)	Slovenia, Austria, Germany	<i>Campylobacter</i> sp.	Biotechnical and Veterinary faculties at the University of Ljubljana, Slovenia, the National Institute of Public Health (NLZOH), Slovenia, the Austrian Agency for Health and Food Safety (AGES), the Institute of Milk Hygiene, Milk Technology and Food Science (Vetmeduni Vienna), Austria, and the Federal Institute for Risk Assessment (BfR), Germany	2002–2014	No
Berthenet et al. (2019)	France	<i>Campylobacter</i> spp.	Structured sampling of <i>Campylobacter</i> in the clinic and from the potential sources during the study	2014–2016	No
Mughini-Gras et al. (2019a, b)	Netherlands (Kingdom of the)	ESBL-producing <i>E. coli</i>	Peer-reviewed publications reporting ESBL and pAmpC gene data in the Netherlands during 2000–15 was systematically reviewed; National Antimicrobial Surveillance System (MARAN) provided additional data	2005–2017	95 % uncertainty interval
Thépault et al. (2018)	France	<i>Campylobacter</i> sp.	Published data and 156 novel <i>C. jejuni</i> genomes sequenced in this study	2009; 2015	95 % uncertainty interval
Lake et al. (2021)	New Zealand	<i>Campylobacter</i> sp.	Notified cases of campylobacteriosis, sampled in different areas of New Zealand during this case-control study; WGS database	2018–2019	No
McLure et al. (2022)	Australia	<i>Salmonella</i> spp.	Treating pathology laboratories reporting <i>Salmonella</i> infections to the health department; data for non-human sources were collated from the National Enteric Pathogens Surveillance System (NEPSS) and the New South Wales Food Authority (NSWFA)	2017–2019	95 % uncertainty interval
Pires and Christensen (2017)	Denmark	<i>Campylobacter</i> sp.	Food monitoring (Danish Veterinary and Food Administration) and Public Health Surveillance (Statens Serum Institute)	2015–2017	95 % uncertainty interval
Mughini-Gras et al. (2021)	Netherlands (Kingdom of the)	<i>Campylobacter</i> sp.	The framework of established surveillance programs for zoonotic agents including <i>Campylobacter</i> ; routine diagnostic activities for human isolates from gastroenteritis patients	2014–2019	95 % uncertainty interval
Mughini-Gras et al. (2016)	Netherlands (Kingdom of the)	<i>Salmonella</i> spp.	National surveillance; Dutch veterinary services (food-producing animals) and private clinics (reptile pets)	1985–2014	95 % uncertainty interval
Mughini-Gras et al. (2014a, b)	Netherlands (Kingdom of the)	<i>Salmonella</i> sp.	EU and national reference laboratory for <i>Salmonella</i> at the Dutch National Institute for Public Health and the Environment (RIVM)	2005–2013	95 % uncertainty interval
Mughini-Gras et al. (2020)	Netherlands (Kingdom of the)	<i>Salmonella</i> sp.	National surveillance data for human samples; Dutch veterinary services (livestock) and private clinics (reptiles)	2005–2019	No
McLure et al. (2023)	Australia	<i>Campylobacter</i> sp.	<i>CampySource</i> project, a collaboration between Australian academic institutions, government agencies, and industry	2017–2019	95 % uncertainty interval
Ahlstrom et al. (2017)	USA	<i>Salmonella</i> spp.	Minnesota Department of Health (MDH); Minnesota Veterinary Diagnostic Laboratory (MVDL)	2005–2014	No
Barco et al. (2015)	Italy	<i>Salmonella</i> spp.	“Enter-net Italia” (human cases); “Enter-Vet Italia” (non-human cases)	2009–2011	95 % uncertainty interval
Hsu et al. (2020)	USA	<i>Campylobacter</i> sp.	National Antimicrobial Resistance Monitoring System;	2013–2017	No
Jabin et al. (2019)	Germany	<i>Salmonella</i> sp.	<i>Salmonella</i> control programs in poultry, coordinated by the BfR on behalf of the German government in year 2010; Public Health Surveillance (Robert Koch Institute (RKI))	2010–2011	95 % uncertainty interval
Merlotti et al. (2020)	Denmark	<i>Salmonella</i> sp., <i>Listeria monocytogenes</i>	Public Health Surveillance (Statens Serum Institute); non-human isolates from Denmark, Germany, Ireland, United Kingdom, and others	2013–2014	95 % uncertainty interval
Perestrelo et al. (2022)	Germany	ESBL-producing <i>E. coli</i>	Fecal, dust and environmental samples from farms with healthy livestock animal; National Reference Laboratory for Antimicrobial Resistance at the Federal Institute for Risk Assessment (BfR) in Berlin, Germany; Institute of Hygiene and Infectious Diseases of Animals (IHIT) in Giessen; National Reference Centre at	2011–2013	No

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Table 2 (continued)

Author (year)	Target population	Pathogen	Data sources (method)	Time period for data collection	Statistical uncertainty provided
Rosner et al. (2017)	Germany	<i>Campylobacter</i> spp.	the Robert Koch Institute (RKI) in Wernigerode, Germany	2011–2014	No
Saif et al. (2022)	Egypt	<i>Campylobacter</i> sp.	German MLST data and available MLST data from neighboring European countries; human samples from the present case-control study	2017–2018	95 % uncertainty interval
Teixeira et al. (2022)	Canada	<i>Campylobacter</i> sp.	Isolates collected during the study; the clinical isolates were recovered from two different hospitals in Cairo, Egypt	2004–2018	95 % uncertainty interval
Hudson et al. (2021)	USA	<i>Campylobacter</i> spp.	Southwestern Alberta (SWA) database	2017–2018	No
Tanui et al. (2022)	All available	<i>Listeria monocytogenes</i>	University of Pennsylvania Health System facilities; metadata from 2013 to 2021	N/A	No
Arning et al. (2021)	All available	<i>Campylobacter</i> sp.	National Centers for Biotechnology Information's (NCBI) Pathogen Detection database from food and clinical sources as part of integrated surveillance	N/A	Yes
Munck et al. (2020)	Denmark	<i>Salmonella</i> spp.	PubMLST (https://pubmlst.org/)	2013–2014	95 % uncertainty interval
Mikkilä et al. (2019)	Finland	<i>Salmonella</i> spp.	Food, animal, and human isolates collected specifically for this study	2010–2015	95 % uncertainty interval
Sapp et al. (2022)	Ethiopia	<i>ETEC</i> , <i>Salmonella</i> spp., <i>STEC</i>	Finnish National Infectious Diseases Register (2018)	2022	95 % uncertainty interval
Sapp et al. (2022)	Burkina Faso	<i>ETEC</i> , <i>Salmonella</i> spp., <i>Campylobacter</i> sp.	NA (Expert elicitation)	2022	95 % uncertainty interval
Sapp et al. (2022)	Rwanda	<i>Campylobacter</i> sp., <i>Salmonella</i> spp., <i>Cryptosporidium</i> sp., <i>Brucella</i> sp., <i>Mycobacterium bovis</i>	NA (Expert elicitation)	2022	95 % uncertainty interval
Butler et al. (2015)	Canada	<i>Bacillus cereus</i> , <i>Brucella</i> sp., <i>Campylobacter</i> sp., <i>Clostridium botulinum</i> , <i>Clostridium perfringens</i> , <i>Cryptosporidium</i> sp., <i>Cyclospora</i> sp., <i>E. coli</i> , <i>ETEC</i> , <i>Giardia</i> sp., hepatitis A, <i>Listeria monocytogenes</i> , norovirus, rotavirus, <i>Salmonella</i> sp., <i>Shigella</i> sp., <i>Staphylococcus aureus</i> , <i>Toxoplasma gondii</i> , <i>Trichinella</i> sp., <i>Vibrio parahaemolyticus</i> , <i>Vibrio</i> sp., <i>Yersinia enterocolitica</i>	NA (Expert elicitation)		95 % uncertainty interval
Beshearse et al. (2021)	USA	<i>Brucella</i> sp., <i>Campylobacter</i> sp., <i>Cryptosporidium</i> sp., <i>E. coli</i> , <i>ETEC</i> , <i>ETEC</i> , <i>Giardia</i> sp., <i>Salmonella</i> sp., <i>Shigella</i> sp., <i>Staphylococcus aureus</i> , <i>Trichinella</i> sp., <i>Vibrio</i> sp., <i>Yersinia enterocolitica</i> , <i>Legionella</i> sp., <i>Streptococcus</i> group A, <i>Pseudomonas</i> sp., <i>Mycobacterium</i> sp.	NA (Expert elicitation)		No
Mughini-Gras et al. (2022)	All available	<i>Campylobacter</i> spp., <i>Salmonella</i> spp., <i>STEC</i> , <i>Listeria monocytogenes</i>	Literature search	2022	No
Mossong et al. (2016)	Luxembourg	<i>Campylobacter</i> sp.	A nationwide case-control study of risk factors for human campylobacteriosis, Grand Duchy of Luxembourg; animal data from Netherlands, Belgium, Luxembourg and France	2010–2013	95 % uncertainty interval
Torgerson et al. (2020)	All available	<i>Cystic and Alveolar echinococcus</i>	Literature search	2020	95 % uncertainty interval
Pintar et al. (2017)	Canada	<i>Campylobacter</i> sp.	Public Health Agency of Canada. FoodNet Canada Healthy Control Survey raw data, 2014; Public Health Agency of Canada. Foodbook, 2014; others	2010–2014	None
Deng et al. (2020)	China	<i>Toxoplasma gondii</i>	Key laboratory of China Food Safety Risk Assessment, Total diet study 2011	2020	None
Christidis et al. (2020)	Canada	<i>Salmonella</i> sp.	Consumption data: Foodbook survey data file and the cycle 2.2 Canadian Community Health Survey (CCHS) data file (Public Health Agency of Canada, 2014b; Statistics Canada, 2004); Public Health Agency of Canada's Foodbook survey; Ontario Ministry of the Environment Data, Ontario Parks data, and a literature review	2020	None

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Table 2 (continued)

Author (year)	Target population	Pathogen	Data sources (method)	Time period for data collection	Statistical uncertainty provided
Ravel et al. (2017)	Canada	<i>Campylobacter</i> sp.	Region of Waterloo (ROW) sentinel site, FoodNet Canada	2017	None
Osbyer et al. (2016)	Cambodia	<i>Campylobacter</i> sp.	Collection of the samples from 3 different regions of Cambodia during the study	2016	N/A
Mughini-Gras et al. (2018)	Netherlands (Kingdom of the)	STEC	National surveillance data	2017	95 % uncertainty interval
Nohra et al. (2020)	New Zealand	<i>Campylobacter</i> sp.	Samples collected during the intervention	2005–2015	95 % uncertainty interval
Hald et al. (2016)	All available	<i>Bacillus cereus</i> , <i>Brucella</i> sp., <i>Campylobacter</i> sp., <i>Clostridium botulinum</i> , <i>Clostridium perfringens</i> , <i>Cryptosporidium</i> sp., <i>Cyclospora</i> sp., <i>E. coli</i> , <i>ETEC</i> , <i>Giardia</i> sp., hepatitis A, <i>Listeria monocytogenes</i> , norovirus, rotavirus, <i>Salmonella</i> sp., <i>Shigella</i> sp., <i>Staphylococcus aureus</i> , <i>Toxoplasma gondii</i> , <i>Trichinella</i> sp., <i>Vibrio parahaemolyticus</i> , <i>Vibrio</i> sp., <i>Yersinia enterocolitica</i>	NA (Expert elicitation)		95 % uncertainty interval
Hoffmann et al. (2017)	All available	<i>Bacillus cereus</i> , <i>Brucella</i> sp., <i>Campylobacter</i> sp., <i>Clostridium botulinum</i> , <i>Clostridium perfringens</i> , <i>Cryptosporidium</i> sp., <i>Cyclospora</i> sp., <i>E. coli</i> , <i>ETEC</i> , <i>Giardia</i> sp., hepatitis A, <i>Listeria monocytogenes</i> , norovirus, rotavirus, <i>Salmonella</i> sp., <i>Shigella</i> sp., <i>Staphylococcus aureus</i> , <i>Toxoplasma gondii</i> , <i>Trichinella</i> sp., <i>Vibrio parahaemolyticus</i> , <i>Vibrio</i> sp., <i>Yersinia enterocolitica</i>	NA (Expert elicitation)		95 % uncertainty interval

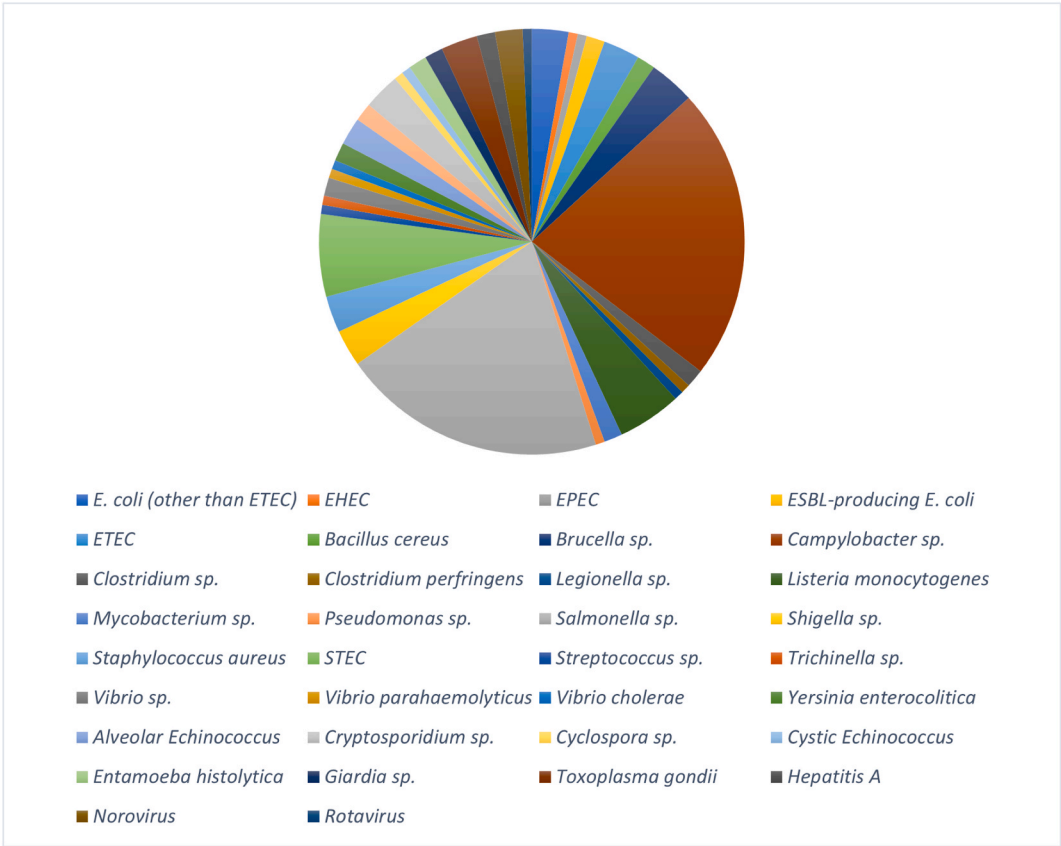


Fig. 2. Overview of the proportion of source attribution studies conducted by pathogen relative to all studies identified (all settings and approaches).

regions and pathogens. A large proportion of the national studies were conducted in Western European, North American, and Western Pacific countries, while attribution estimates from the remaining regions were largely lacking. This lack of studies in non-Western countries highlights the need for further data collection and research in many parts of the world in the long term, as well as the use of non-data-driven methods for

source attribution of foodborne pathogens in the short term. Moreover, while there were various studies covering the same few pathogens (e.g. *Salmonella*, *Campylobacter*), or the same country, there were very few studies for other pathogens. For example, only one study focused on rotavirus, *Echinococcus*, *Pseudomonas* sp., and *Clostridium* spp., and only two studies looked at *Bacillus cereus*, *Entamoeba histolytica*, and hepatitis A. The studies filling some of these gaps were analyses of data from outbreak investigations, which covered a range of pathogens at once. We categorized studies by WHO region, but food exposure routes may vary substantially across subregions. For example, the first round of WHO global burden of foodborne disease estimates found substantial differences in food sources across WHO's North America subregion and its two subregions in Latin America (Hald et al. (2016); Hoffmann et al. (2017)).

Pires et al. (2014) reviewed *Salmonella* source attribution studies published up to 2012 and compared methodologies and estimates across studies. The authors concluded that the differences in data and methods contributed to considerable variability in estimates across the studies. While analyzing estimates by pathogen and country or region was beyond the scope of our review, lessons can be drawn from this review of the available evidence on source attribution globally. Importantly, additional research estimating the relative contribution of transmission pathways and sources at the regional and global levels is crucial to provide a better understanding of the global burden and source attribution of foodborne pathogens. The expert elicitation study conducted under the first estimation of the global burden of foodborne diseases by the WHO (WHO 2015) (Aspinall et al. (2016); Hald et al. (2016); Hoffmann et al. (2017)) represented the first effort to fill in this gap. The study-specific source attribution estimates we collected in this review will help inform a new source attribution study based on expert elicitations conducted in the context of FERG for 2021–2025 activities.

This review has limitations. The literature search was conducted only in PubMed. Although we have taken steps to reduce the risk of missing relevant articles by conducting additional consultations with experts of FERG for 2021–2025 and scanning the reference lists of included articles, we cannot exclude the possibility of having missed studies. The literature search was conducted in 2023, and only studies published between January 2010 and July 2023 were included. Studies published after this review, e.g. Guzinski et al. (2024) and Hurst et al. (2023), which both attributed salmonellosis, were not captured.

Knowing which foods are associated with a particular foodborne disease is essential to develop food safety interventions targeted at the most important sources. The presence of similarities and differences in relative contributions of different food sources and routes of transmission in different countries shows regional differences in epidemiology of various foodborne diseases.

Research on source attribution methods has advanced markedly in the past 20 years. Previous reviews have described available source attribution methods, their strengths and limitations, and applicability of source attribution approaches for different pathogens (Mughini-Gras et al. (2018), Mughini-Gras et al. (2019a,b), Pires et al. (2009); Wagenaar et al. (2013)). Our review reveals geographic gaps and opportunities for the application of additional methods to strengthen the evidence on source attribution. Additional work at the national and regional levels all contribute to improving our knowledge of the epidemiology of pathogens globally.

CRedit authorship contribution statement

Aleksandra Davydova: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation. **Christina Fastl:** Writing – review & editing, Validation, Methodology. **Lapo Mughini-Gras:** Writing – review & editing. **Li Bai:** Writing – review & editing. **Kunihiro Kubota:** Writing – review & editing. **Sandra Hoffmann:** Writing – review & editing, Validation, Data curation. **Tety Rachmawati:** Writing – review & editing. **Sara M. Pires:**

Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization.

Data statement

Data supporting this study is available through the Zenodo repository <https://doi.org/10.5281/zenodo.14720777> under the Creative Commons Attribution 4.0 International license.

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Declaration of competing interest

The authors declare that they have no known competing financial interests that could have influenced the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.fm.2025.104812>.

Data availability

I have shared a link to my data at the Attach file step
Extracted data from 62 published source attribution studies, 2010-2023 and database of all the studies included in the review Source attribution studies of foodborne pathogens 2010-2023 (Reference data) (Zenodo)

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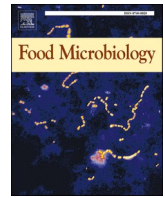
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Corrigendum



Corrigendum to ‘Source attribution studies of foodborne pathogens, 2010–2023: a review and collection of estimates’ [Food Microbiol., Volume 131/October 2025, 104812]

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The correct affiliation of Tety Rachmawati is “Research Center for Public Health and Nutrition, National Research and Innovation Agency,

Indonesia”, replacing originally indicated “Center of Research and Development for Humanities and Health Management, Indonesia”.

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